

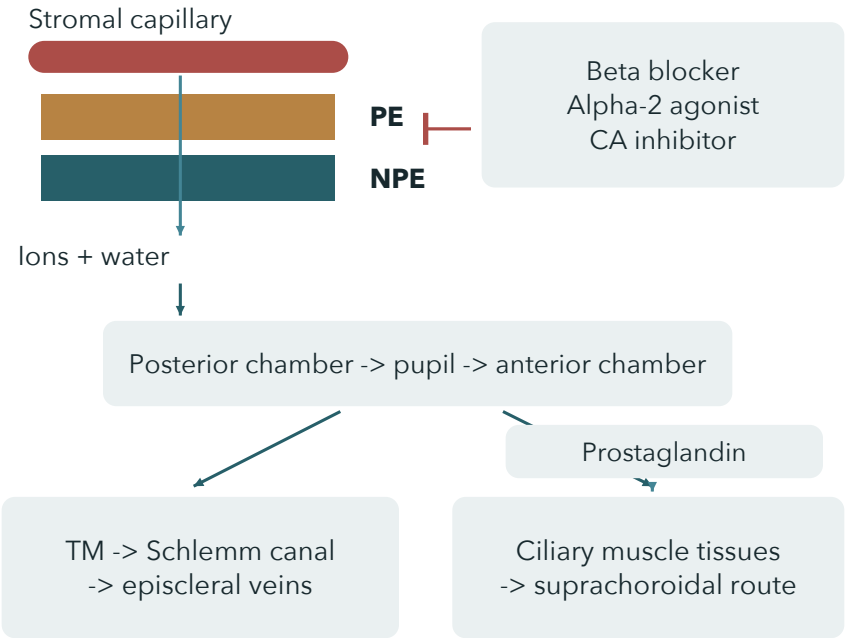
Glaucoma drug mechanisms

TYPE B | Follow the target, then predict the flow change.

Formation or drainage?

One aqueous circuit, several targets

FORMATION: CILIARY EPITHELIUM



CONVENTIONAL

UVEOSCLERAL

PE = pigmented epithelium; NPE = nonpigmented epithelium; CA = carbonic anhydrase. Medicines change formation or outflow without necessarily changing angle anatomy.

Target → response

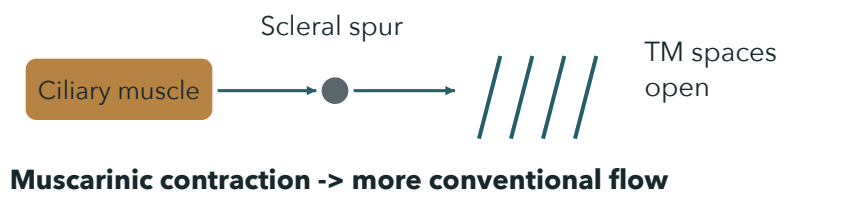
Class / example	Main high-yield effect
Beta blocker / timolol	Less aqueous formation
Alpha-2 agonist / brimonidine	Less formation; more uveoscleral outflow
CA inhibitor / acetazolamide, dorzolamide	Less epithelial bicarbonate-linked secretion
Prostaglandin analogue / latanoprost	More uveoscleral outflow

Flow checkpoint

$IOP = \frac{\text{episcleral venous pressure} + (\text{formation} - \text{uveoscleral flow})}{\text{conventional facility}}$. If all other terms are fixed, less formation or more facility lowers IOP.

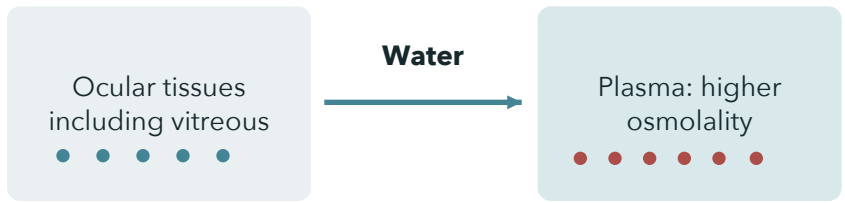
Mechanical and osmotic effects

Two different routes to a lower pressure PILOCARPINE: MECHANICAL LINK



Muscarinic contraction -> more conventional flow

MANNITOL: OSMOTIC WATER SHIFT



Ocular water/volume falls while barriers remain intact

Pilocarpine changes conventional drainage through a mechanical linkage.
Mannitol redistributes water by an osmotic gradient.

Mechanism and connected effect

Drug	Connected consequence
Timolol	Systemic beta blockade can cause bradycardia or bronchospasm
Acetazolamide	Renal bicarbonate loss can cause metabolic acidosis
Pilocarpine	Miosis and accommodation-related blur or brow ache
Mannitol	Ocular water moves towards a more concentrated plasma compartment

Do not merge the mechanisms

A smaller pupil does not prove that aqueous production fell.
Prostaglandin analogues act mainly through unconventional outflow; pilocarpine acts through muscarinic ciliary contraction and trabecular drainage.